BTS – IV – 04.14 – 0107

B

B.Tech. Degree IV Semester Examination April 2014

ME 404 APPLIED THERMODYNAMICS
(*2006 Scheme*)

Time : 3 Hours Maximum Marks : 100

PART A
(Answer *ALL* questions)

(8 × 5 = 40)

I. (a) State and prove Carnot's theorem.
(b) Explain the term 'availability'.
(c) What is 'triple point'? Give the pressure and temperature of water at its triple point.
(d) Explain the basic components of a steam power plant.
(e) State Dalton's law of partial pressures.
(f) Differentiate gravimetric and volumetric analysis.
(g) What is meant by stoichiometric air-fuel ratio?
(h) Define the terms (i) enthalpy of formation (ii) absolute or standardized enthalpy.

PART B

(4 × 15 = 60)

II. (a) A reversible engine operates between 27°C and 327°C. What is the least rate of heat rejection per kW net output of the engine? (6)
(b) State the Kelvin Planck and Clausius statements of second law of thermodynamics. Also show the equivalence of the two statements. (9)

OR

III. (a) Establish the inequality of clausius. (6)
(b) $0.04m^3$ of nitrogen contained in a cylinder behind a piston is initially at a state of 1.05 bar and 15°C. The gas is compressed isothermally and reversibly until the pressure is 4.8 bar. Calculate: (9)
(i) the entropy change
(ii) the heat flow
(iii) the work done
Sketch the process on a $p-v$ and $T-S$ diagram. Assume nitrogen as a perfect gas.
Molecular weight of nitrogen = 28

IV. (a) Describe a reheat cycle and compare it with simple Rankine cycle. (8)
(b) In a steam power cycle, the steam supply is at 15 bar and dry saturated. The condenser pressure is 0.4 bar. Calculate the Carnot and Rankine efficiencies of the cycle. Neglect pump work. (7)

OR

V. (a) Discuss the desirable characteristics of a working fluid in a vapor power cycle. (6)
(b) A steam power station uses the following cycle: (9)

Steam at boiler outlet	–	150 bar, 550°C
Reheat at	–	40 bar to 550°C
Condenser at	–	0.1 bar

Using the Mollier chart and assuming ideal processes, find the (i) quality at turbine exhaust (ii) cycle efficiency (iii) steam rate

(P.T.O.)

VI.	(a)	A vessel of $0.35m^3$ capacity contains 0.4kg of carbonmonoxide (molecular weight = 28) and 1 kg of air at 20°C. Calculate: (i) The partial pressure of each constituent (ii) The total pressure in the vessel The gravimetric analysis of air is to be taken as23.3% oxygen (molecular weight = 32) and 76.7% nitrogen (molecular weight = 28)	(8)
	(b)	Briefly explain the working of a rotary compressor with neat diagram.	(7)
		OR	
VII.	(a)	A single stage single acting reciprocating air compressor has air entering at 1 bar, 20°C and compression occurs following polytropic process with index 1.2 upto the delivery pressure 12 bar. The compressor runs at the speed of 240rpm and has L/D ratio of 1.8. The compressor has mechanical efficiency of 0.88. Determine the isothermal efficiency and cylinder dimensions. Also find out the rating of drive required to run the compressor which admits $1m^3$ of air per minute.	(9)
	(b)	Express Amagat's law of additive volumes. Does this law hold exactly for ideal gas mixtures?	(6)
VIII.	(a)	Propane (C_3H_8) is burned with 75 percent excess air during a combustion process. Assuming complete combustion, determine the air-fuel ratio on both a mass basis and a molar basis.	(6)
	(b)	Explain the adiabatic flame temperature in constant pressure and constant volume combustion processes.	(9)
		OR	
IX.	(a)	The gravimetric analysis of a sample of coal is given as 80% C, 12% H_2 and 8% ash. Calculate the stoichiometric air-fuel ratio and the analysis of the products by volume.	(8)
	(b)	Describe the method for the determination of calorific values of a solid fuel with neat sketch.	(7)

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